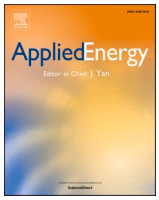
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Physics-informed neural network for long-term prognostics of proton exchange membrane fuel cells

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HIGHLIGHTS

- PINN was innovatively developed to predict the remaining useful life of PEMFC.
- Prior knowledge of PEMFC degradation and system variability was integrated into PINN.
- PINN exhibited high prediction accuracy and low dependency on input data length.

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ABSTRACT

This study formulated a physics-informed neural network (PINN) to prognosticate the remaining useful life (RUL) of proton exchange membrane fuel cell (PEMFC), leveraging both prior knowledge of PEMFCs and aging test data. Governing equations elucidating membrane and catalyst degradation mechanisms were integrated into the PINN framework to assimilate the prior understanding of PEMFC degradation dynamics. Subsequently, operation time and current density were designated as input variables for the PINN to forecast output voltage, following a comprehensive analysis of PEMFC operational mechanisms and aging test data. Notably, a novel strategy involving replicating the current density from the training phase to the prediction phase was introduced, incorporating prior knowledge of PEMFC system variability into the PINN. Consequently, the proposed PINN demonstrated proficient prediction of PEMFC RUL while mitigating reliance on aging test data. This accomplishment, representing a 9.2 percentage points enhancement over the previously lowest reported data dependency of 35.2 %, substantiates the attainment of state-of-the-art status by the proposed PINN in reducing data dependency.

1. Introduction

Proton exchange membrane fuel cells (PEMFCs) have garnered considerable attention as promising sustainable power generation devices [1] owing to their high energy efficiency [2], high power density [3] and lack of harmful emissions during operation [4]. However, their short lifespan substantially hinders large-scale commercialization [5]. Therefore, extensive research has been conducted on prognosis and health management (PHM) to extend their longevity. PHM entails prognosticating the remaining useful life (RUL) of a PEMFC and undertaking proactive maintenance [6] or operation parameter feedback control [7] based on the predicted RUL. Precise RUL prediction is paramount because the efficacy of PHM in ensuring PEMFC longevity relies on accurately predicting the RUL [8].

A model-based approach has been explored to achieve accurate RUL prediction. The approach expresses the impact of PEMFC degradation on its performance and lifespan through governing equations derived from prior physical knowledge of PEMFC operating mechanisms [9]. The accuracy of RUL estimation in this approach significantly improves when the governing equations incorporate precise prior knowledge that aptly describes PEMFC degradation mechanisms. However, a challenge arises owing to the intricate interactions among components, such as the membrane [10], catalyst [11], and electrode [12], which contribute to actual PEMFC degradation. This systematic degradation, not extensively

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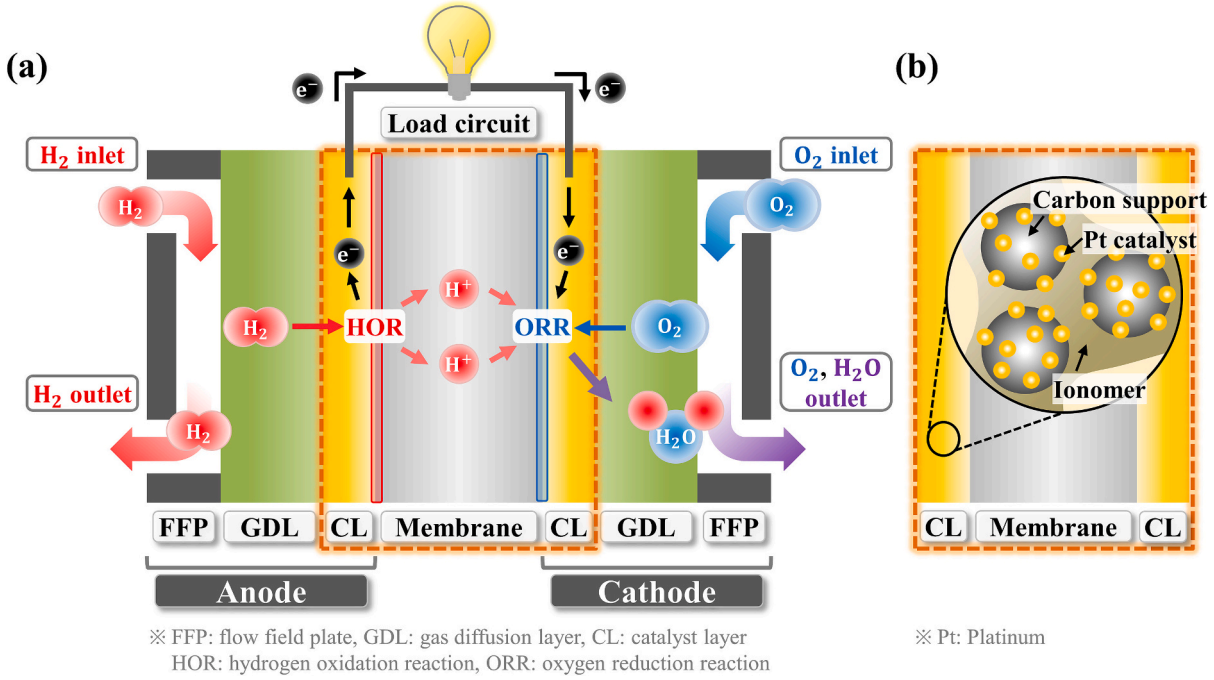


Fig. 1. Operating mechanism of a PEMFC. (a) a schematic representation of the electrochemical reactions occurring in different parts of the PEMFC. (b) an internal structure of the catalyst layers adjacent to the membrane (Note: Image not to scale).

studied in theoretical investigations, raises uncertainty about the feasibility of developing a single governing equation capable of elucidating every complex degradation process [13]. Hence, the model-based approach encounters limitations in estimating the RUL of a PEMFC under standard operating conditions characterized by complex degradation processes [14].

As an alternative, data-driven approaches have garnered increasing interest owing to advancements in artificial intelligence. Unlike the model-based approach, the data-driven approach obviates the need for prior knowledge of the complex systematic degradation mechanism of PEMFC. The data-driven approach utilizes experimental data acquired from aging tests as input for a deep learning (DL) model, training it to estimate the RUL [15]. DL model in the data-driven approach can circumvent the necessity for prior knowledge because it can directly discern the degradation trend of data resulting from the complex degradation of a PEMFC system [16]. Consequently, data-driven approaches can surmount the limitations of model-based approaches. In the data-driven approach, the DL model is trained to approximate the degradation trend by minimizing the error between the aging test data and predicted values, necessitating a substantial volume of training data [17]. However, the challenge arises owing to the large-scale commercialization of PEMFC not being achieved and the costly nature of conducting aging tests, making it difficult to obtain sufficient aging test data [18]. Furthermore, inadequate data for training DL models can lead to issues such as overfitting and divergence of the error value, ultimately compromising the reliability and accuracy of the model [19]. Hence, ongoing research efforts are focused on developing DL model structures capable of effectively mitigating the data dependency concerns associated with DL models. One such effort involves introducing the input data length (IDL) as an evaluation index. The IDL is defined as the percentage of input data relative to the entire lifetime of the system. A reduction in the IDL required by the DL model to forecast the RUL indicates improved effectiveness in terms of data dependency [20]. Previous research in this field includes the work of Zhang et al. [21], who developed a DL model structure using a long short-term memory (LSTM) layer. This model effectively considers historical voltage fluctuation characteristics during the forecasting phase because voltage data can reflect the degradation trend of PEMFC. Consequently, when voltage data representing 61.8 % IDL were used to train the DL model, a remarkable 95 % prediction accuracy of RUL was achieved. Khaled et al. [22] devised a DL model structure employing a convolutional neural network layer, effectively capturing the cyclic rise and fall patterns observed in aging data by focusing on the recovery phenomenon in PEMFC during periodic start-up and shut-down cycles. Consequently, using voltage data corresponding to 57.5 % IDL for the DL model, they attained an RUL prediction accuracy of 83 %. Hua et al. [23] introduced a multi-input DL model designed to concurrently consider the effects of additional sensor data measured during the aging test on the output voltage data because the sensor data influenced the RUL prediction result. The prediction accuracy of PEMFC RUL reached 97.3 % by incorporating voltage, hydrogen input temperature, and hydrogen input pressure, representing 51 % IDL. The minimum IDL value reported for the data-driven approach remained at 51 % despite advancements in developing DL models with reduced data dependence. It is attributed to the diminishing potential inadequacy of the dataset in capturing sufficient degradation-related information to effectively train the DL model. This limitation, stemming from the reliance on available data to capture complex degradation processes, presents a fundamental obstacle to data-driven approaches that cannot be readily overcome.

Therefore, a hybrid approach was introduced to estimate the RUL accurately, even with limited input data. [24]. This strategy aims to diminish reliance on the quantity of aging data required for training the DL model while integrating prior knowledge encapsulated in governing Eqs. [25]. Among various hybrid methods, physics-informed neural network (PINN) have recently gained considerable traction across diverse fields of study. PINN stands out owing to its exceptional prediction performance, achieved by directly incorporating governing equations into the DL model training process [26]. The architecture of PINN uniquely facilitates simultaneous training on prior knowledge embedded within governing equations and observed degradation trends by including a loss term that quantifies the disparity between outcomes predicted by the governing equations and those generated by the DL model. Consequently, PINN can mitigate the dependence of the DL model on the quantity of aging test data by incorporating governing

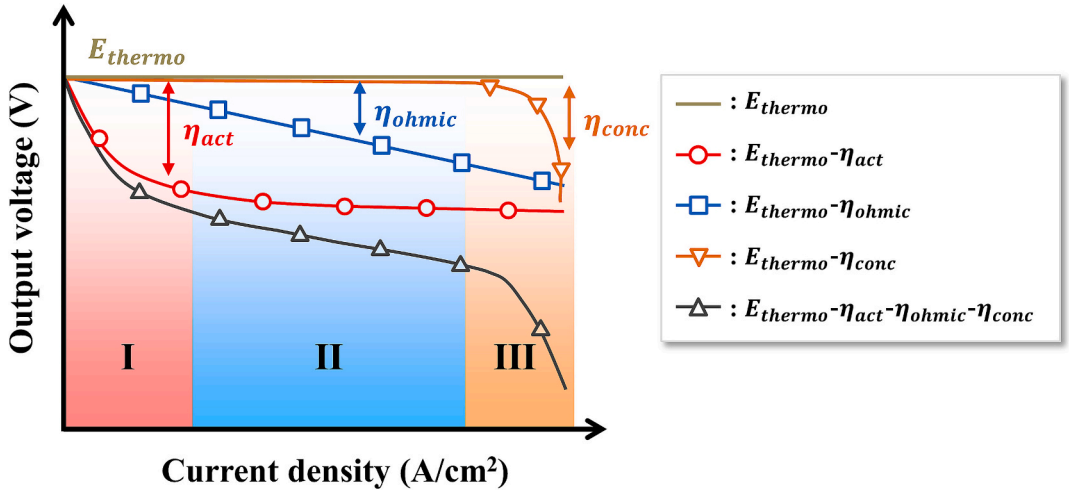


Fig. 2. Polarization curve of the PEMFC, showcasing the trends in activation loss (η_{act}), ohmic loss (η_{ohmic}), and concentration loss (η_{conc}) concerning current density.

equations embodying prior knowledge. Therefore, PINN can alleviate the data dependence of the DL model, which trains degradation trends from aging data by integrating these governing equations into the training process of the DL model. However, to the best of our knowledge, no studies have yet reported the application of PINN to predict the RUL of PEMFC [27].

Hence, this study employed PINN for the first time to predict the RUL of PEMFC. Using this newly developed PINN, the required IDL for RUL prediction was notably reduced to 26 %, pioneering a method to decrease dependency on aging test data. To develop a PINN for this innovative application, aging data and prior knowledge should be acquired, and consequently, a suitable DL model structure and governing equations for the PINN should be determined. First, the open dataset from the "IEEE PHM Data Challenge" long-term test dataset [28] was chosen to assess the prediction performance of the PINN. Subsequently, the input variable necessary for accurate RUL prediction of PEMFC was deduced by understanding prior knowledge regarding the electrochemical mechanisms of PEMFC. Especially, a novel method for generating current density as an input variable for the PINN in the prediction phase was employed. Additionally, governing equations explaining prior knowledge of membrane and catalyst degradation trends were incorporated. A PINN capable of estimating PEMFC RUL was developed for the first time by applying these governing equations. The adequacy of prior knowledge incorporated into the PINN model was evaluated by comparing RUL prediction accuracy before and after applying prior knowledge. Finally, we determined whether the PINN achieved state-of-the-art (SOTA) status by comparing the prediction accuracy of the PINN with that of the hybrid approach, which reported the lowest IDL in the literature [29].

2. Operating mechanism of PEMFC

As stated in the Introduction, developing a PINN for predicting the RUL of PEMFC requires selecting and integrating prior knowledge to augment information regarding the degradation mechanism into the DL model. Consequently, in this section, we introduce an operating mechanism of PEMFC to elucidate its basic domain knowledge [30]. PEMFC generates water, electricity, and heat through electrochemical reactions at the anode and cathode (Fig. 1(a)). The hydrogen oxidation reaction (HOR) and oxygen reduction reaction (ORR) occur at the interface between the membrane and the catalyst layer for anode and cathode, respectively. The chemical reaction equations for the HOR and ORR are presented in Eqs. (1) and (2), respectively, with the overall process described by Eq. (3).

$$\text{HOR at the anode}: H_2 \rightarrow 2H^+ + 2e^- \quad (1)$$

$$\text{ORR at the cathode}: \frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O \quad (2)$$

$$\text{Overall cell reaction}: \frac{1}{2}O_2 + H_2 = H_2O + electricity + heat \quad (3)$$

The electrochemical reactions in the PEMFC process occur in the following order: First, a gaseous hydrogen molecule (H_2) at the anode enters through the H_2 inlet of the flow-field plate and moves through the gas diffusion layer. Subsequently, the hydrogen molecule interacts with the platinum (Pt) catalyst supported by carbon support (Fig. 1(b)). The chemical bonds between hydrogen atoms are broken during this interaction, forming two separate hydrogen ions (H^+) and two electrons. Ionomers provide ion conductivity, facilitating the movement of ions within the catalyst layer [37]. The dissociated hydrogen ions then move through the membrane, while the electrons are transported to the catalyst layer in the cathode through the external load circuit. Hydrogen molecules that do not participate in the electrochemical reaction are emitted through H_2 outlet of the flow-field plate.

Oxygen molecules are supplied at the cathode through O_2 inlet of the flow-field plate and pass through the gas diffusion layer to reach the catalyst layer. The oxygen molecules react with the electrons traveling through the load circuit and the hydrogen ions passing through the membrane. This reaction produces water, and any oxygen molecules that do not participate in the electrochemical reaction are released through the outlet of the flow-field plate. An ideal thermodynamic voltage (E_{thermo}) is generated in the load circuit through these series of processes. However, the actual measured output voltage of the PEMFC is always lower than the theoretically calculated E_{thermo} primarily because of three types of voltage losses that inevitably occur during the electrochemical reaction, categorized as activation, ohmic, and concentration losses. Understanding the underlying mechanisms behind these losses and their effects on the output voltage is crucial because they can significantly influence the output voltage.

The causes of each type of loss and their effects on the output voltage are summarized as follows.

1. **Activation loss (η_{act})**: η_{act} refers to the voltage loss occurring at the interface between the catalyst layer and the membrane, where hydrogen and oxygen molecules are dissociated into their ionic states. As illustrated by the red solid line with a circular marker ($E_{thermo} - \eta_{act}$) in Fig. 2, which depicts the polarization curve of the PEMFC, the trend in η_{act} varies across different current density

Table 1
Controlled operating conditions of tested PEMFC 1 and PEMFC 2.

Controlled operating conditions	Control range
Cooling temperature [°C]	20–80
Cooling flow [*l*/*min*]	0–10
Gas temperature [°C]	20–80
Gas humidification [% RH]	0–100
Airflow [*l*/*min*]	0–100
H_2 flow [*l*/*min*]	0–30
Gas pressure [bars]	0–2
Fuel cell current [A]	0–300

regions. In the low current density region (region I), η_{act} increases with increasing current density and follows an inverse exponential relationship, with a diminishing increase across regions II and III. This behavior is attributed to the potential difference created by the hydrogen ions in the membrane and electrons in the catalyst layer at the interface, which can dissociate newly supplied hydrogen molecules without additional voltage loss in regions II and III.

2. **Ohmic loss (η_{ohmic})**: η_{ohmic} represents the voltage loss from the internal resistance generated because dissociated hydrogen ions are transported from the anode through the membrane toward the cathode. As illustrated by the blue solid line with a rectangular marker ($E_{thermo} - \eta_{ohmic}$) in Fig. 2, η_{ohmic} showcases a linear increase with the rise in current density across regions I, II, and III. This linear relationship can be explained using Ohm's law: η_{ohmic} increases linearly with increasing current density because the resistance is regarded as a constant value determined by the ion conductivity, area, and length of the membrane.
3. **Concentration loss (η_{conct})**: η_{conc} refers to the voltage loss occurring when excessive hydrogen and oxygen molecules are supplied to increase the current density. As depicted by the orange solid line with an inverted triangular marker ($E_{thermo} - \eta_{conc}$) in Fig. 2, η_{conc} is primarily observed in the high current density region (region III) because, unlike in regions I and II, excessive hydrogen and oxygen molecules that do not participate in the electrochemical reaction and remain undissociated may obstruct the process of supplying new fuel and expelling reaction products.

The polarization curve of the output voltage (V) can be represented by the black solid line with a triangular marker in Fig. 2 by incorporating the effect of the aforementioned losses ($\eta_{act}, \eta_{ohmic}, \eta_{conc}$) on E_{thermo}. The polarization curve can be expressed using Eq. (4)

$$V = E_{thermo} - \eta_{act} - \eta_{ohmic} - \eta_{conc} \tag{4}$$

3. Experimental setup of open dataset

This section outlines the experimental setup of the open dataset and the aging test data used to train the DL model of the PINN. The Fuel Cell

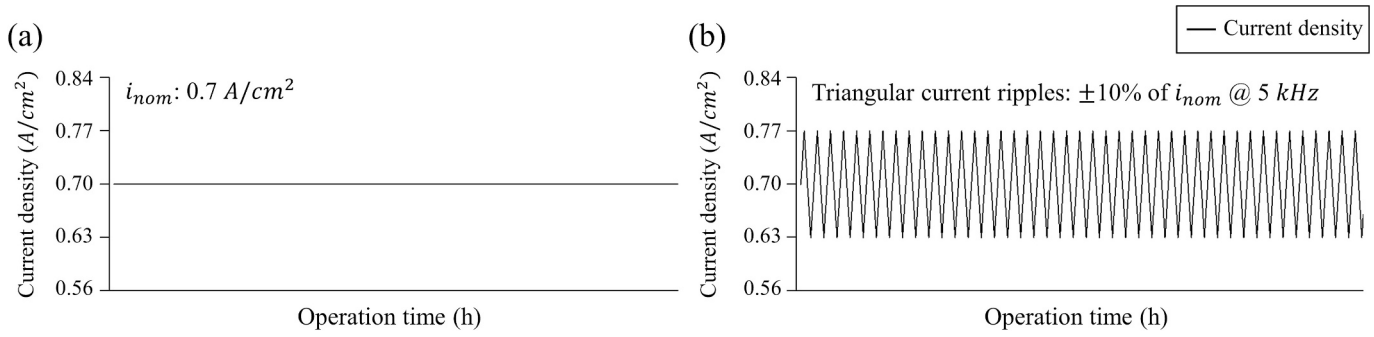


Fig. 3. Current density profiles for PEMFC 1 and PEMFC 2. (a) Nominal current density (i_{nom}: 0.7 A/cm^2) applied to PEMFC 1 and (b) triangular current ripples (±10% of i_{nom} at a frequency of 5 kHz) applied to PEMFC 2.

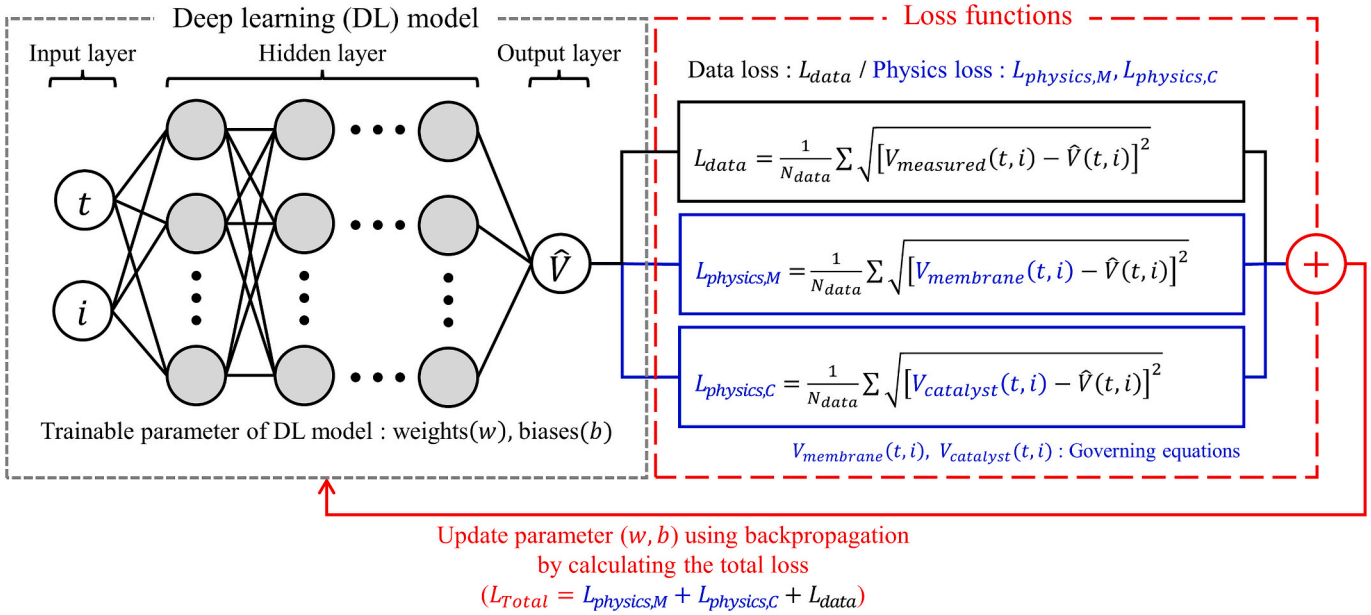


Fig. 4. Structure of PINN used to predict the output voltage of the PEMFC.

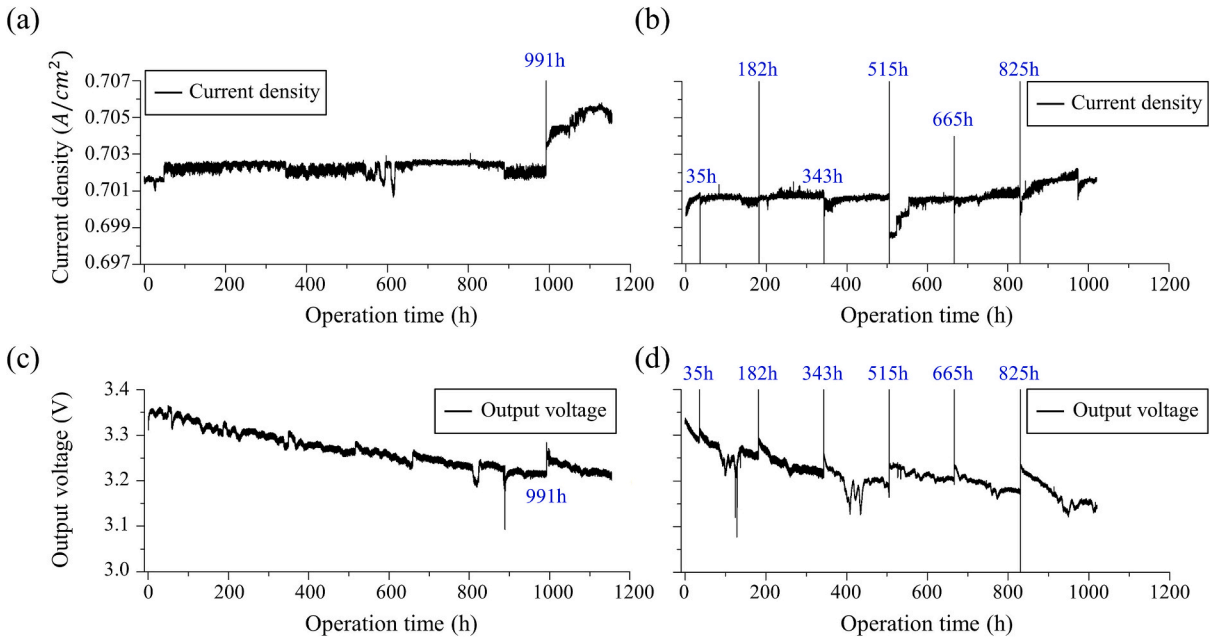


Fig. 5. Measured current density and output voltage for PEMFC 1 and PEMFC 2 over the operation time. Current density for (a) PEMFC 1 and (b) PEMFC 2. Output voltage for (c) PEMFC 1 and (d) PEMFC 2.

Laboratory (FCLAB) Research Federation (FR CNRS 3539, France, http: fclab.fr) collected and compiled the aging test data. The aging test involved using a PEMFC stack comprising five cells with an active area of 100 cm^2. The operating conditions of the PEMFC were maintained within specified ranges to ensure the stability of the aging test (Table 1). Moreover, two aging tests with distinct current density profiles were conducted for two sets of data (PEMFC 1 and PEMFC 2) to analyze the impact of degradation trend differences regarding the loading conditions (Fig. 3).

Fig. 3 illustrates the current density profiles applied to PEMFC 1 and PEMFC 2. In the case of PEMFC 1 (Fig. 3(a)), the preset value of the nominal current density (i_{nom}) value remained constant at 0.7 A/cm^2 throughout the test. Conversely, the preset value of the current density for PEMFC 2 (Fig. 3(b)) fluctuated in a triangular ripple at a frequency of 5 kHz, ranging between 0.63 A/cm^2 and 0.77 A/cm^2. This range corresponds to a ± 10 % deviation from the i_{nom}.

4. Methodology

4.1. Proposed physics-informed neural network (PINN)

In this section, we introduce the PINN used for predicting the output voltage of the PEMFC, followed by the proposed method for estimating the RUL. Section 4.1 elucidates the theoretical background of the PINN and outlines the structure of the PINN developed in this study. A PINN is a model structure designed to integrate governing equations reflecting prior knowledge into the DL model when approximating a function representing the overall trend of the data based on the universal approximation theorem [31]. In a typical DL model, which lacks integration of governing equations, data loss is calculated by comparing predicted values with the actual values, followed by backpropagation to minimize data loss by updating the trainable parameters—the weights and biases. The physical loss for a PINN was additionally calculated using the governing equation to apply it to the backpropagation process. Consequently, the PINN considers both data and physical losses arising from the discrepancy between the predicted value and the governing equation when calculating the loss. This integration enables the PINN to effectively train from both the data and the governing equations, facilitating the approximation of a function that accurately represents the degradation trend of the aging test data.

The structure of the proposed PINN for predicting the output voltage of a PEMFC based on the theoretical background of the PINN is depicted in Fig. 4. The DL model within the PINN, represented inside the dotted gray box, uses the operation time and current density as input variables to predict the output voltage. The internal structure of the DL model adopts that of a deep neural network (DNN) [32]. The rationale for incorporating the current density and output voltage as input variables for the DNN is explained in Section 4.2. The DNN used to construct the PINN comprised of input, hidden, and output layers. The input layer was composed of two nodes, each representing operation time and current density as input variables, respectively. The hidden layer comprised of eight layers, each containing 20 nodes, and we utilized the hyperbolic tangent (tanh) as an activation function. The output layer consisted of a single node that predicted the output voltage.

The loss functions used to compute the data and physical losses are depicted by the dashed red box in Fig. 4. These loss functions comprised one data loss (L_{data}) and two distinct types of physics losses ($L_{physics,m}$, $L_{physics,c}$). These losses sum to the total loss (L_{total}). Specifically, the two types of physical losses calculate the difference between the values computed from the governing equations, each of which characterizes the degradation of the catalyst and membrane. The reasons for selecting such governing equations and their detailed descriptions are explained in Section 4.3.

The input data underwent two distinct training stages during the PINN training process. The primary difference between each training phase was the learning rate required to effectively achieve the global minimum rather than the local minimum during the backpropagation process. During the model training process, the Adam (Adaptive Moment Estimation) algorithm was used as the optimizer for adjusting the weights and biases [33]. In the first training stage, the PINN was trained with a learning rate of 10^{-3} for 5000 epochs. Subsequently, the learning rate in the second training stage was reduced to 10^{-6} for an additional 5000 epochs. The loss used for backpropagation throughout each training stage was calculated as the root mean square error (RMSE), with the total loss calculated by adding the data and physics losses.

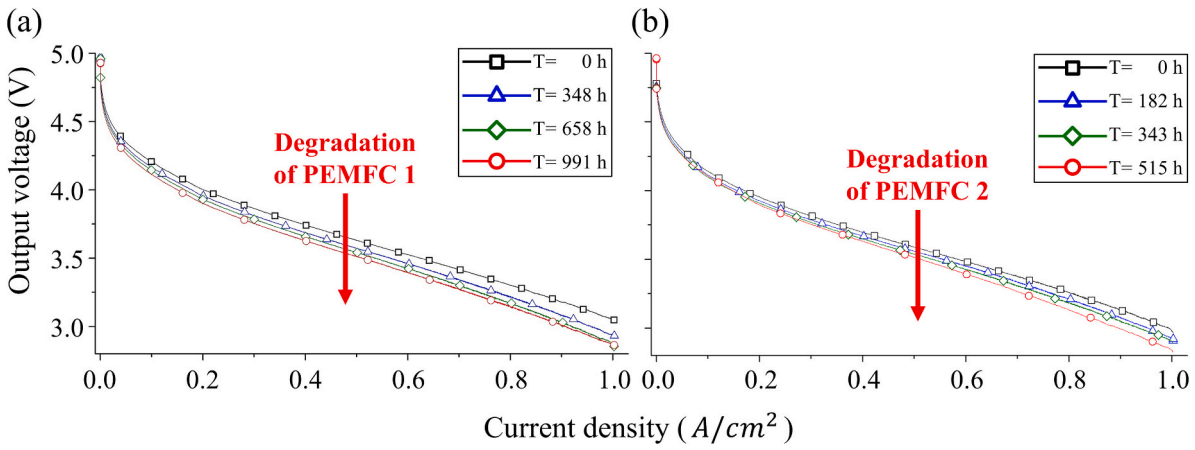


Fig. 6. Change in polarization curve concerning the degradation in the PEMFC. (a) PEMFC 1 and (b) PEMFC 2.

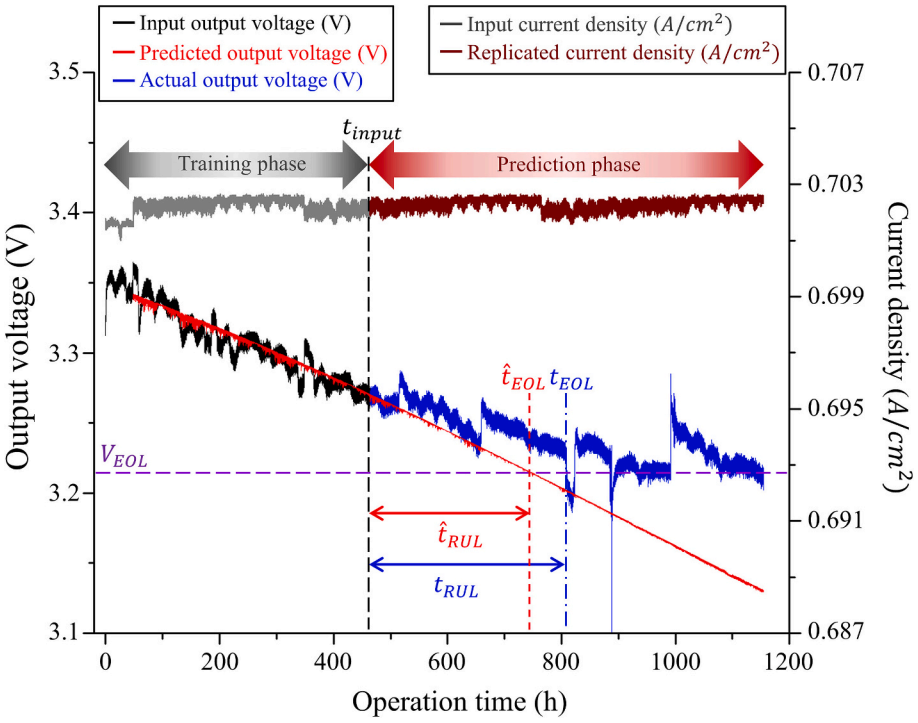


Fig. 7. A comprehensive result of PEMFC RUL prediction steps by using the output voltage predicted from PINN with operation time and current density used as input variables.

4.2. Selecting input variables for voltage prediction

Section 4.2 presents the theoretical and experimental background for selecting the aforementioned input variables of the PINN to predict the output voltage (introduced in Section 4.1), based on the analysis of the aging test data. First, the distinct aging tests conducted on PEMFC 1 and PEMFC 2 under the experimental conditions outlined in Section 3 were analyzed. The measured output voltages and current densities obtained from these tests are illustrated in Fig. 5. A critical observation from analyzing this time-series format of the aging test data is the presence of uncontrollable variabilities stemming from the inability to precisely regulate the current density profile (Fig. 3) to achieve the targeted output voltage and power. These variabilities are illustrated in Fig. 5(a) and (b), respectively. This is due to the variability associated with the control of current density in actual PEMFC power generation mechanisms [34]. Such variability arises because controlling the current density of PEMFCs requires precise regulation of various operational conditions, including air and hydrogen stoichiometry, air-hydrogen inlet pressure, temperature, and flow rates [35]. However, these operational conditions fluctuate during PEMFC operation due to factors such as heat generation and turbulence in hydrogen and oxygen supply, which inevitably cause variations in current density and affect power generation [36]. They can be categorized into two types: small oscillations in the data and significant fluctuations, particularly at 991 h for PEMFC 1 and at 35, 182, 343, 515, 665, and 825 h for PEMFC 2. These current density variabilities ultimately influence the output voltage level, as depicted in Fig. 5(c) and (d). Therefore, the output voltage of the PEMFC showed a declining trend as degradation occurs, accompanied by variability owing to changes in the current density.

The polarization curve (detailed in Section 2) elucidates the influence of changes in current density on the output voltage [37]. Fig. 6(a) and (b) depict the polarization curves plotted during the aging tests of PEMFC 1 and PEMFC 2, respectively. These curves demonstrate that the output voltage decreases across all current densities with increasing

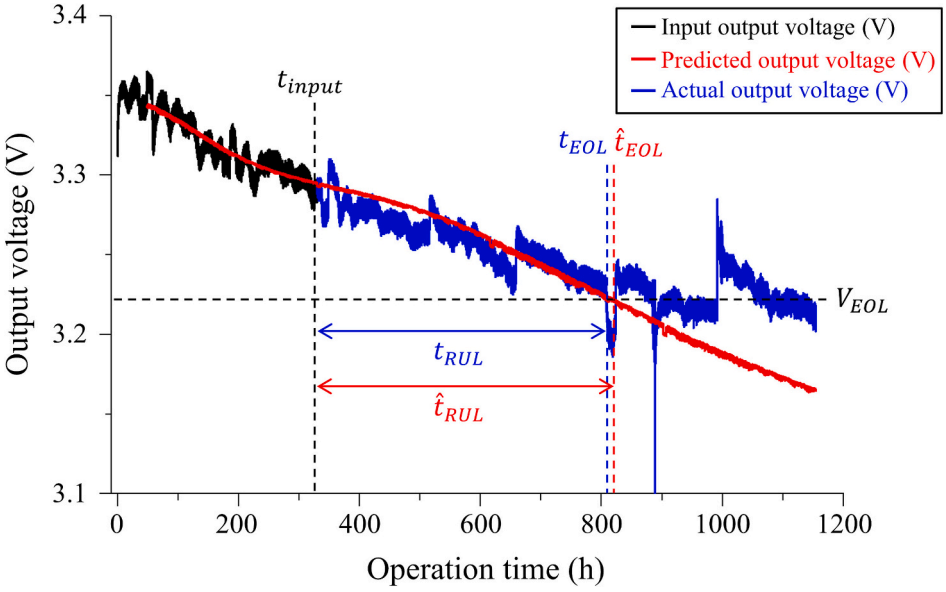


Fig. 8. Plot of the predicted output voltage over operation time for PEMFC 1 at an IDL value of 40.8 % (330 h).

Table 2
Comparison of the RUL prediction accuracy and error of the PINN for PEMFC 1 at IDL 40.8 % (330 h) based on the applicability of current density and governing equations.

Model	Mean RUL prediction accuracy	Mean error	Standard deviation
PINN without governing equations	68.25 %	31.75 %	28.36 %
PINN without current density	92.99 %	−7.01 %	0.47 %
Proposed PINN	**95.21 %**	**−4.79 %**	**0.58 %**

operation time. This decline originates from the irreversible increase in voltage loss due to component degradation, such as membrane, catalyst, within the PEMFC over time, thereby resulting in a reduced output voltage at the same current density. Therefore, the overall downward trend in output voltage is attributable to PEMFC degradation by examining the output voltages of PEMFC 1 and PEMFC 2 over time. Furthermore, the recurrent variabilities in the output voltage can be attributed to variations in current density. Consequently, operation time and current density were chosen as input variables for the PINN during voltage prediction, exerting the most significant influence on the output voltage.

4.3. Selecting governing equations for the loss function

As elucidated in Section 4.1, two governing equations were used to

Table 3
Comparison of the RUL prediction accuracy and error of the PINN for PEMFC 2 at an IDL of 35.2 % (330 h) based on the applicability of current density and governing equations.

Model	RUL prediction accuracy	Mean error	Standard deviation
PINN without governing equations	Not converged		
PINN without current density	98.03 %	1.97 %	0.16 %
Proposed PINN	**99.53 %**	**−0.47 %**	**0.29 %**

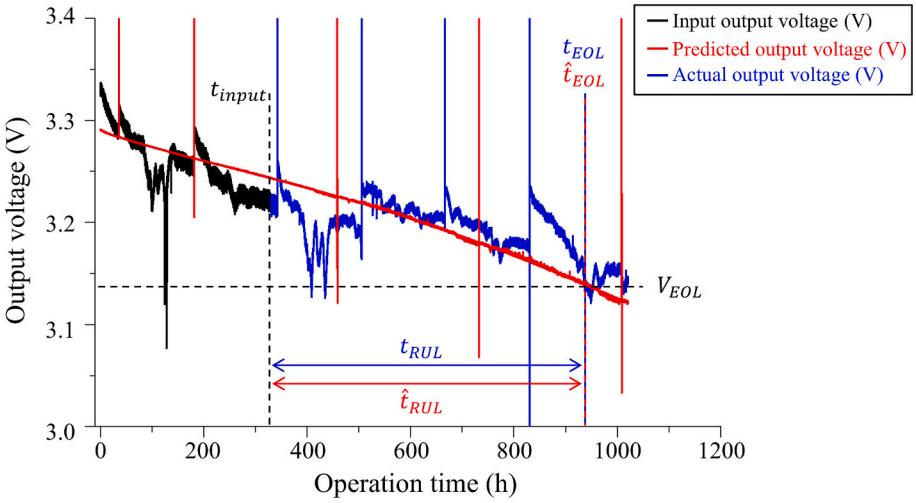


Fig. 9. Plot of the predicted output voltage over operation time for PEMFC 2 at an IDL value of 35.2 % (330 h).

Table 4
Results comparing RUL prediction accuracy and error for PEMFC 1 at IDL values of 26 % (210*h*) and 40.8 % (330 h) between the SOTA and the proposed PINN.

Model	210 h (26 %)		330 h (40.8 %)	
	RUL prediction accuracy	Error	RUL prediction accuracy	Error
SOTA	–	–	94 %	6 %
Proposed PINN	**91.3 %**	**8.7 %**	**95.9 %**	**−4.1 %**

compute the physical loss and model catalyst and membrane degradation, respectively. First, the governing equations must inherently consider these variables while predicting the output voltage because the DL model approximates the output voltage based on operation time and current density. Hence, this study adopted governing equations from previous research that modeled the effect of PEMFC degradation on output voltage while considering operation time and current density as independent variables. These studies conducted accelerated life tests under conditions inducing local membrane and catalyst degradation, vital components significantly influencing PEMFC performance. Subsequently, they developed governing equations modeling the relationship between output voltage, current density, and operating time [38]. Therefore, the following two governing equations each representing degradation of membrane and catalyst, which are components that significantly affect the PEMFC degradation, were applied to the PINN to compute the physical loss within the loss function:

The first governing equation, developed by Han et al., delineates the influence of membrane degradation on the output voltage [39]. This equation identifies membrane water content as the primary factor contributing to output voltage decline. As membrane degradation progresses, water content increases, diminishing ohmic conductivity within the membrane and consequently increasing ohmic loss [40]. Hence, a governing equation was derived based on accelerated-life test results to induce membrane degradation. This equation incorporated operation time as a variable affecting the water-content indicator to calculate the ohmic loss. The governing equation is expressed in Eqs. (49), (53), and (56) in [39]. At this stage, the governing equations incorporate key parameters such as the PEMFC's operating temperature, membrane thickness, and hydrogen and oxygen inlet pressures. These parameters enable the PINN model to effectively integrate information regarding the PEMFC's structural characteristics and operating conditions.

The second governing equation, formulated by Fowler et al., characterizes the effect of catalyst degradation on output voltage [41]. This equation identifies the actual catalyst surface area reduction as the primary factor leading to output voltage decline. Catalytic activity diminished owing to decreased surface area, resulting in increased activation loss. Hence, a governing equation was developed based on accelerated life test findings to induce catalyst degradation. This equation incorporates operation time as a variable affecting the indicator representing the actual catalyst surface area in the equation used to calculate activation loss. The specific equations are shown in Eqs. (1)–(7) in [41]. Since the governing equations also considers the membrane thickness and its area, the reaction temperature of the PEMFC, as well as the inlet and exit flow rates and pressures, information about the PEMFC's structure and operating conditions can be incorporated into the PINN.

4.4. RUL prediction of PEMFC by using PINN

The RUL prediction of the PEMFC by using PINN is processed in the following steps.

1. Training/prediction phase designation step: The operational time range containing all PEMFC input data was divided into two phases (Fig. 7). The phase encompassing data up to t_{input}, marking the end of input aging test data, was designated as the training phase. The period beyond t_{input} was designated as the prediction phase. This division facilitated training the PINN with measured aging test data during the training phase and using it for output voltage prediction during the prediction phase.

Table 5
Results comparing the RUL prediction accuracy and error for PEMFC 2 at IDL values of 26 % (244 h) and 35.2 % (330 h) between the SOTA and the proposed PINN.

Model	244 h (26 %)		330 h (35.2 %)	
	RUL prediction accuracy	Error	RUL prediction accuracy	Error
SOTA	–	–	97.2 %	−2.8 %
Proposed PINN	**95.6 %**	**4.4 %**	**99.9 %**	**−0.1 %**

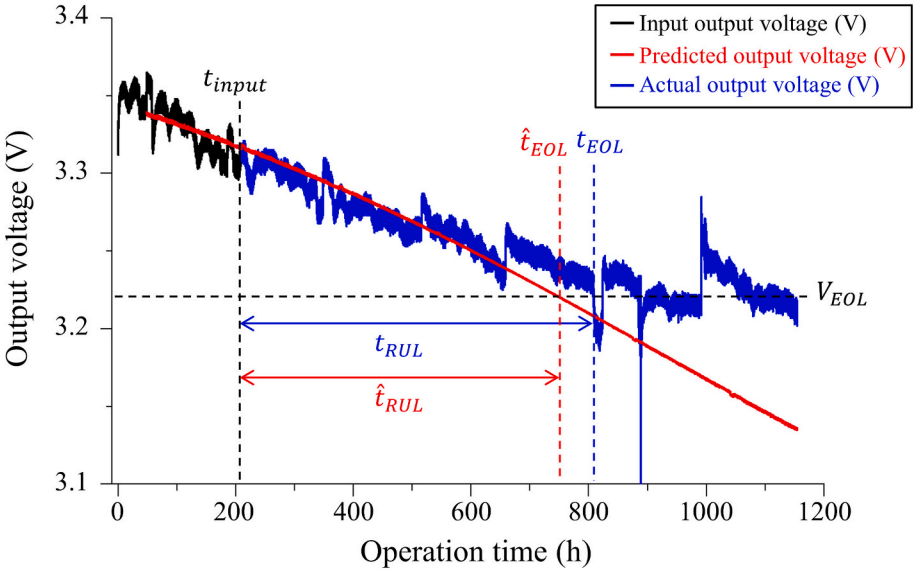


Fig. 10. Predicted output voltage plotted against operation time for PEMFC 1 at an IDL value of 26 % (210 h).

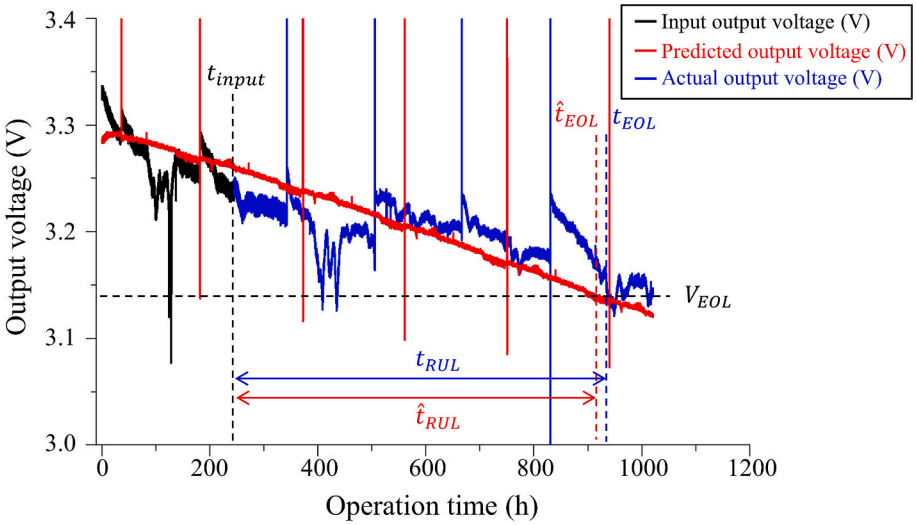


Fig. 11. Predicted output voltage plotted against operation time for PEMFC 2 at an IDL value of 26 % (244 h).

2. PINN training step: The PINN was trained (as explained in Section 4.1) using aging test data from the training phase determined in the previous step.
3. Output voltage prediction step: Current density data required as input variables for the previously trained PINN were generated for the prediction phase, and the output voltages in the prediction phase were predicted using the generated current density. In this step, the current density data for the prediction phase were generated because the PINN requires this as input to forecast the output voltage. Especially, current density data replication was necessary because current density, unaffected directly by PEMFC degradation (Section 4.2), relied on user-defined values and system variability. Hence, a novel method was developed to replicate current density data from the time of maximum voltage measurement to t_{input}. Current density for the prediction phase was generated by duplicating the current density data at the training phase (Fig. 7). Current density at the prediction phase served as input to predict the output voltage in the prediction phase.
4. RUL calculation step: The RUL of the PEMFC was calculated based on the predicted output voltage obtained using the PINN. This calculation was performed because PEMFC failure is determined when its output voltage decreases by a predefined percentage (e.g., 10 %) [42]. The RUL calculation index can be observed in Fig. 8 [43]. Initially, V_{EOL} (Fig. 7, purple dashed line) represents the predetermined output voltage level defining the PEMFC failure point. The time taken for the predicted output voltage to reach V_{EOL} is denoted as $\hat{t}_{EOL}$ (Fig. 7, red dotted line), representing the expected lifespan of the PEMFC. Subsequently, the RUL from the conclusion of the aging test (t_{input}) to the point of failure was determined as $\hat{t}_{RUL}$, which denotes the difference between $\hat{t}_{EOL}$ and t_{input}.

4.5. Performance evaluation index for the RUL prediction model

The RUL prediction process (Section 4.4) evaluated the performance of the PINN using two key indexes: RUL prediction error and RUL prediction accuracy, which were calculated using Eqs. (5) and (6) (known as evaluation indices).

$$\text{RUL prediction error } (\%) = \frac{t_{RUL} - \hat{t}_{RUL}}{t_{RUL}} \times 100 \tag{5}$$

$$\text{RUL prediction accuracy } (\%) = \left(1 - \frac{|t_{RUL} - \hat{t}_{RUL}|}{t_{RUL}}\right) \times 100 \tag{6}$$

Both indexes serve to compare the actual RUL (t_{RUL}) with the predicted RUL, providing insights into the performance of the model. Therefore, actual RUL was determined by the difference between the end of the lifetime of the PEMFC (t_{EOL}) and the input time (t_{input}), as illustrated in Fig. 7. An acceptable error range of −8 % to 16 % was defined based on recommendations by Jouin et al. [44] to evaluate the success of the model in RUL prediction. This range facilitates sufficient time for maintenance and repair before the complete failure of the PEMFC.

Additionally, the input data length (IDL) was introduced to assess the dependence of the model on input data. IDL, defined by Eq. (7), represents the percentage of input data relative to the lifetime of PEMFC.

$$IDL\ (\%) = \frac{t_{input}}{t_{EOL}} \times 100 \tag{7}$$

This study uses the minimum IDL value necessary to achieve RUL prediction within the acceptable error range; providing insight into the data dependency of the model in RUL prediction.

5. Results and discussion

5.1. Experimental validation of proposed PINN

Section 4 outlines a novel technique for the newly developed PINN and its application in predicting and evaluating the RUL of a PEMFC. Section 5.1 discusses RUL predictions conducted on the aging test data presented in Section 3 to validate the RUL prediction performance of the PINN. In this process, the RUL prediction accuracy was compared based on integrating key characteristics into the PINN, such as using current density as an input variable and incorporating governing equations to calculate the physical loss. The IDL used for RUL prediction sets at 40.8 % (330*h*), aligning with the approach used in the SOTA publication [29].

Fig. 8 displays a plot of the voltage predicted using the proposed PINN (red solid line) for PEMFC 1, alongside the input voltage (black solid line) and the actual voltage (blue solid line), plotted over the operation time. The RUL prediction accuracy of the proposed PINN (Table 2) indicates an accuracy of 95.21 % at an IDL of 40.8 % (330 h). Comparative analysis of RUL predictions were conducted to assess the influence of integrating the current density and governing equations—the principal attributes of PINN—on the accuracy of RUL predictions, each omitting one of these characteristics. The experiment involved performing 20 repeated predictions for each case to analyze the distribution of prediction errors, and compare the mean and standard deviation of the prediction accuracy. As explained earlier, applying

governing equations to the PINN representes incorporating prior knowledge of PEMFC degradation into the DL model. As described in Table 2, when prior knowledge of PEMFC degradation was not applied by omitting the governing equations, the RUL prediction accuracy droped from 95.21 % to 68.25 %, indicating a decrease of 26.96 percentage points. Furthermore, using current density as an input variable in the PINN, as described in Section 4.2, enables the incorporation of the prior knowledge regarding the PEMFC variability into the model. As shown in the PINN results without current density in Table 2, which is a condition where the current density was not used as an input variable, the RUL prediction accuracy decreased from 95.21 % to 92.99 %, indicating a reduction of 2.22 percentage points, and the standard deviation decreased from 0.58 % to 0.47 %, a drop of 0.11 percentage points, due to the exclusion of current density variability.

Fig. 9 depicts a graphical plot for PEMFC 2 illustrating the output voltage predicted by the proposed PINN with a red solid line, the input voltage represented by a black solid line, and the actual voltage shown by a blue solid line, all plotted with respect to the operation time. As shown in Fig. 9, at the same 35.2 % IDL (330 h) as the SOTA, PINN achieved an RUL prediction accuracy of 99.53 %. Similar to PEMFC 1, Table 3 presents a comparative RUL prediction experiment to verify the performance of the proposed PINN by examining the impact of current density and governing equations, which are the key features of the PINN. The comparative RUL prediction experiment, as in the previous case, involved performing 20 repeated predictions for each case to analyze the distribution of prediction errors, and compare the mean and standard deviation of the prediction accuracy. First, as shown in Table 3, the results of PINN without governing equations, where prior knowledge of PEMFC degradation was not applied, led to divergent prediction results because of insufficient training data, ultimately resulting in failure in RUL prediction. Next, the results of PINN without current density showed that, since prior knowledge of PEMFC variability was not incorporated by omitting current density, the RUL prediction accuracy decreased from 99.53 % to 98.03 %, indicating a reduction of 1.5 percentage points. The standard deviation also decreased from 0.29 % to 0.16 %, as the variability of PEMFC was not included. Through the experimental validation of the proposed PINN conducted on PEMFCs 1 and 2, the effectiveness of incorporating prior knowledge of PEMFC degradation and variability into the PINN in the forms of governing equations and current density, respectively, in improving RUL prediction accuracy has been confirmed.

5.2. Comparing RUL prediction accuracy with that of previous study

Following the validation of the PINN in Section 5.1, a comparative analysis in Section 5.2 aimed to highlight the superiority of the PINN by comparing it with the SOTA, which had previously established the benchmark for the most successful RUL prediction at the lowest IDL. The RUL prediction results for PEMFC 1 are summarized in Table 4. According to the SOTA publication, a RUL prediction accuracy of 94 % was achieved with a 6 % margin of error at an IDL of 4.8 % (330 h). As depicted in Table 2, the proposed PINN demonstrated enhanced prediction performance with an accuracy of 95.9 % for an identical IDL, surpassing the SOTA confirming the superior prediction performance of the PINN. Furthermore, the RUL prediction process depicted in Fig. 10 and its summarized outcomes in Table 4 show that the PINN successfully forecasted the RUL with 91.3 % accuracy at an IDL of 26 % (210 h). This result indicates that the proposed PINN can achieve high RUL prediction accuracy even with only 26 % of PEMFC aging data. This percentage was 14.8 % lower than the accuracy achieved by the SOTA model, demonstrating the enhanced performance of the PINN in reducing the data dependency.

The RUL prediction results for PEMFC 2 across various IDL values are summarized in Table 5. As per the previously mentioned SOTA, the RUL was successfully predicted with a − 2.8 % error margin at an IDL of 35.2 % (330 h), resulting in an RUL prediction accuracy of 97.2 %. A comparison with the data presented in Table 3 reveals why the proposed PINN achieves a significantly higher accuracy of 99.9 % in forecasting the RUL of the PEMFC 2. Additionally, the output voltage prediction plot depicted in Fig. 11 and the summarized results in Table 5 reaffirm that the proposed PINN can precisely predict the RUL with a 95.6 % precision at an IDL of 26 % (244 h). This demonstrates the ability of the PINN to accurately forecast the RUL using only 26 % of the total aging data of the PEMFC, even amid fluctuations in current density. Notably, this IDL value is 9.2 percentage points lower than 35.2 % (330 h) held by the previous SOTA model. These findings provide compelling evidence of the enhanced data dependency of the PINN.

6. Conclusions

This study represents a pioneering effort in developing a PINN tailored specifically for predicting the RUL of PEMFC by directly integrating prior knowledge concerning PEMFC degradation into the DL model. The proposed PINN is adept at assimilating the degradation characteristics of PEMFC and the system-level variability alongside the aging test data. First, the prior knowledge regarding the degradation mechanism of PEMFC was learned by the PINN by applying governing equations, which model the degradation characteristics of key components of PEMFC, such as the membrane and catalyst, to the loss functions of PINN. This resulted in an improvement of RUL prediction accuracy by 26.96 % and 99.53 % points (from diverged to converged) for the two types of data, respectively. Next, the prior knowledge of PEMFC system variability was learned by the PINN by using current density, which determines the variability occurring at the PEMFC system level, as an input variable for predicting output voltage. This resulted in an improvement of RUL prediction accuracy by 2.22 % and 1.5 % points for the two data, respectively. As a result, in terms of data dependence, the minimum IDL required for acceptable RUL prediction decreased from 35.2 % in the existing SOTA model to 26 %, underscoring a reduction of 9.2 % points. Through this achievement, the effect of reducing data dependence in DL models by incorporating prior knowledge has been confirmed. When applying the proposed PINN to real-world cases, it is expected that more noise and outliers will be present when compared to lab cases, and these bottlenecks are expected to negatively impact RUL prediction accuracy. In future work, we intend to develop appropriate data pre-processing techniques and improve the structure of the DL model used for building the PINN to enhance RUL prediction accuracy in real-world cases.

CRediT authorship contribution statement

Taehwan Ko: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Dukyong Kim:** Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Jaewoong Park:** Validation, Investigation. **Seung Hwan Lee:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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Data availability

The authors do not have permission to share data.

References

[1] Li Q, Wang H, Wang T, Li X, Liu Y, Chen W, et al. Online diagnosis method of water management faults based on hybrid-frequency electrochemical impedance spectroscopy for PEMFC. IEEE Transactions on Transportation Electrification 2024: 1. https://doi.org/10.1109/tte.2024.3427401.

[2] Zhao J, Tu Z, Chan SH. Carbon corrosion mechanism and mitigation strategies in a proton exchange membrane fuel cell (PEMFC): a review. J Power Sources 2021: 488. https://doi.org/10.1016/j.jpowsour.2020.229434.

[3] Shen J, Du C, Yan F, Chen B, Tu Z. Experimental study on the dynamic performance of a power system with dual air-cooled PEMFC stacks. Appl Energy 2022:326. https://doi.org/10.1016/j.apenergy.2022.120025.

[4] Oh H, Lee W-Y, Won J, Kim M, Choi Y-Y, Han S-B. Residual-based fault diagnosis for thermal management systems of proton exchange membrane fuel cells. Appl Energy 2020:277. https://doi.org/10.1016/j.apenergy.2020.115568.

[5] Li J, Cheng Y, Yu H, Du H, Cui H. Multi-agent deep meta-reinforcement learning-based active fault tolerant gas supply management system for proton exchange membrane fuel cells. eTransportation 2023:18. https://doi.org/10.1016/j.etran.2023.100282.

[6] Zhang C, Hu H, Ji J, Liu K, Xia X, Nazir MS, et al. An evolutionary stacked generalization model based on deep learning and improved grasshopper optimization algorithm for predicting the remaining useful life of PEMFC. Appl Energy 2023:330. https://doi.org/10.1016/j.apenergy.2022.120333.

[7] Li Q, Su T, Yin L, Xie S, Tan Y, Chen W, et al. Health management for PEMFC system long-term operation based on optimal temperature trajectory real-time optimization. IEEE Trans Ind Electron 2024:1–11. https://doi.org/10.1109/tie.2024.3436539.

[8] Zhang Z, He H, Wang Y, Quan S, Chen J, Han R. A novel generalized prognostic method of proton exchange membrane fuel cell using multi-point estimation under various operating conditions. Appl Energy 2024:357. https://doi.org/10.1016/j.apenergy.2023.122519.

[9] Zhang X, Zhang T, Chen H, Cao Y. A review of online electrochemical diagnostic methods of on-board proton exchange membrane fuel cells. Appl Energy 2021:286. https://doi.org/10.1016/j.apenergy.2021.116481.

[10] Okonkwo PC, Ben Belgacem I, Emori W, Uzoma PC. Nafion degradation mechanisms in proton exchange membrane fuel cell (PEMFC) system: a review. Int J Hydrogen Energy 2021;46:27956–73. https://doi.org/10.1016/j.ijhydene.2021.06.032.

[11] Fan L, Zhao J, Luo X, Tu Z. Comparison of the performance and degradation mechanism of PEMFC with Pt/C and Pt black catalyst. Int J Hydrogen Energy 2022; 47:5418–28. https://doi.org/10.1016/j.ijhydene.2021.11.135.

[12] Okonkwo PC, Ige OO, Barhoumi EM, Uzoma PC, Emori W, Benamor A, et al. Platinum degradation mechanisms in proton exchange membrane fuel cell (PEMFC) system: a review. Int J Hydrogen Energy 2021;46:15850–65. https://doi.org/10.1016/j.ijhydene.2021.02.078.

[13] Chu T, Zhang R, Wang Y, Ou M, Xie M, Shao H, et al. Performance degradation and process engineering of the 10 kW proton exchange membrane fuel cell stack. Energy 2021:219. https://doi.org/10.1016/j.energy.2020.119623.

[14] Ruiz Diaz DF, Wang Y. Performance loss due to gas coverage on catalyst surface in polymer electrolyte membrane electrolysis cell. eTransportation 2023:18. https://doi.org/10.1016/j.etran.2023.100263.

[15] Liu H, Chen J, Hissel D, Lu J, Hou M, Shao Z. Prognostics methods and degradation indexes of proton exchange membrane fuel cells: a review. Renew Sustain Energy Rev 2020:123. https://doi.org/10.1016/j.rser.2020.109721.

[16] Tao Z, Zhang C, Xiong J, Hu H, Ji J, Peng T, et al. Evolutionary gate recurrent unit coupling convolutional neural network and improved manta ray foraging optimization algorithm for performance degradation prediction of PEMFC. Appl Energy 2023;336. https://doi.org/10.1016/j.apenergy.2023.120821.

[17] Zuo J, Lv H, Zhou D, Xue Q, Jin L, Zhou W, et al. Deep learning based prognostic framework towards proton exchange membrane fuel cell for automotive application. Appl Energy 2021:281. https://doi.org/10.1016/j.apenergy.2020.115937.

[18] Wang C, Li Z, Outbib R, Dou M, Zhao D. Symbolic deep learning based prognostics for dynamic operating proton exchange membrane fuel cells. Appl Energy 2022: 305. https://doi.org/10.1016/j.apenergy.2021.117918.

[19] Hong J, Yang H, Liang F, Li K, Zhang X, Zhang H, et al. State of health prediction for proton exchange membrane fuel cells combining semi-empirical model and machine learning. Energy 2024:291. https://doi.org/10.1016/j.energy.2024.130364.

[20] Jacome A, Hissel D, Heiries V, Gerard M, Rosini S. Prognostic methods for proton exchange membrane fuel cell under automotive load cycling: a review. IET Electrical Systems in Transportation 2020;10:369–75. https://doi.org/10.1049/iet-est.2020.0045.

[21] Zhang Z, Wang Y-X, He H, Sun F. A short- and long-term prognostic associating with remaining useful life estimation for proton exchange membrane fuel cell. Appl Energy 2021:304. https://doi.org/10.1016/j.apenergy.2021.117841.

[22] Benaggoune K, Yue M, Jemei S, Zerhouni N. A data-driven method for multi-step-ahead prediction and long-term prognostics of proton exchange membrane fuel cell. Appl Energy 2022:313. https://doi.org/10.1016/j.apenergy.2022.118835.

[23] Hua Z, Zheng Z, Péra M-C, Gao F. Remaining useful life prediction of PEMFC systems based on the multi-input echo state network. Appl Energy 2020:265. https://doi.org/10.1016/j.apenergy.2020.114791.

[24] Wang C, Dou M, Li Z, Outbib R, Zhao D, Liang B. A fusion prognostics strategy for fuel cells operating under dynamic conditions. eTransportation 2022:12. https://doi.org/10.1016/j.etran.2022.100166.

[25] Raissi M, Perdikaris P, Karniadakis GE. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J Comput Phys 2019;378:686–707. https://doi.org/10.1016/j.jcp.2018.10.045.

[26] Kim SW, Kwak E, Kim J-H, Oh K-Y, Lee S. Modeling and prediction of lithium-ion battery thermal runway via multiphysics-informed neural network. Journal of Energy Storage. 2023:60. https://doi.org/10.1016/j.est.2023.106654.

[27] Pombo DV, Rincón MJ, Bacher P, Bindner HW, Spataru SV, Sørensen PE. Assessing stacked physics-informed machine learning models for co-located wind–solar power forecasting. Sustainable Energy, Grids and Networks 2022:32. https://doi.org/10.1016/j.segan.2022.100943.

[28] Research FCLAB. IEEE PHM 2014 data challenge. http://eng.fclab.fr/ieee-phm-2014-data-challenge/; 2014.

[29] Zhou D, Al-Durra A, Zhang K, Ravey A, Gao F. Online remaining useful lifetime prediction of proton exchange membrane fuel cells using a novel robust methodology. J Power Sources 2018;399:314–28. https://doi.org/10.1016/j.jpowsour.2018.06.098.

[30] Makharia R, Kocha S, Yu P, Sweikart MA, Gu W, Wagner F, et al. Durable PEM fuel cell electrode materials: requirements and benchmarking methodologies. Ecs Transactions 2006;1:3.

[31] Cybenko G. Approximation by superpositions of a sigmoidal function. Mathematics of control, signals and systems 1989;2:303–14.

[32] Goh AT. Back-propagation neural networks for modeling complex systems. Artificial intelligence in engineering 1995;9:143–51.

[33] Diederik PK. Adam: A method for stochastic optimization. 3rd International Conference on Learning Representations (ICLR). San Diego, CA, USA. 2014.

[34] Daud WRW, Rosli RE, Majlan EH, Hamid SAA, Mohamed R, Husaini T. PEM fuel cell system control: a review. Renew Energy 2017;113:620–38. https://doi.org/10.1016/j.renene.2017.06.027.

[35] Serra M, Aguado J, Ansede X, Riera J. Controllability analysis of decentralised linear controllers for polymeric fuel cells. J Power Sources 2005;151:93–102. https://doi.org/10.1016/j.jpowsour.2005.02.050.

[36] Meidanshahi V, Karimi G. Dynamic modeling, optimization and control of power density in a PEM fuel cell. Appl Energy 2012;93:98–105. https://doi.org/10.1016/j.apenergy.2011.04.048.

[37] Zhang X, Yang D, Luo M, Dong Z. Load profile based empirical model for the lifetime prediction of an automotive PEM fuel cell. Int J Hydrogen Energy 2017;42: 11868–78. https://doi.org/10.1016/j.ijhydene.2017.02.146.

[38] Mayur M, Strahl S, Husar A, Bessler WG. A multi-timescale modeling methodology for PEMFC performance and durability in a virtual fuel cell car. Int J Hydrogen Energy 2015;40:16466–76. https://doi.org/10.1016/j.ijhydene.2015.09.152.

[39] Han J, Han J, Yu S. Investigation of FCVs durability under driving cycles using a model-based approach. Journal of Energy Storage 2020:27. https://doi.org/10.1016/j.est.2019.101169.

[40] Wang J. System integration, durability and reliability of fuel cells: challenges and solutions. Appl Energy 2017;189:460–79. https://doi.org/10.1016/j.apenergy.2016.12.083.

[41] Fowler MW, Mann RF, Amphlett JC, Peppley BA, Roberge PR. Incorporation of voltage degradation into a generalised steady state electrochemical model for a PEM fuel cell. J Power Sources 2002;106:274–83.

[42] McQueen S, Stanford J, Satyapal S, Miller E, Stetson N, Papageorgopoulos D, et al. Department of energy hydrogen program plan. In: US Department of Energy (USDOE), Washington DC (United States); 2020. https://doi.org/10.1557/adv.2016.495.

[43] Hua Z, Zheng Z, Pahon E, Péra M-C, Gao F. A review on lifetime prediction of proton exchange membrane fuel cells system. J Power Sources 2022:529. https://doi.org/10.1016/j.jpowsour.2022.231256.

[44] Jouin M, Bressel M, Morando S, Gouriveau R, Hissel D, Péra M-C, et al. Estimating the end-of-life of PEM fuel cells: guidelines and metrics. Appl Energy 2016;177: 87–97. https://doi.org/10.1016/j.apenergy.2016.05.076.